



CODE OF PRACTICE
OPS/103/001

Standard for the Numbering of Overhead Lines

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Revision Record

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1. Purpose

The purpose of this Code of Practice is to specify the method of numbering overhead line support in order to ensure consistency throughout the Distribution System.

This Code of Practice supersedes Operation Standard No 1, issued December 1978, all copies of which shall be destroyed.

2. Scope

The method consists of allocating a line number to a section of line and then serially numbering the supports within that line. In the case of LV lines provision is made for each line to be divided into a number of parts and the supports separately numbered within each part.

Numbering shall be introduced as and when new construction work or alterations to the system justify and application to existing system should be carried out at every opportunity. As an interim measure Line Numbers (and where appropriate Part Numbers) will be assigned and used in conjunction with the existing Support Number to enable the supports to be put on the computer record.

3. Numbering System for HV Lines

3.1 Definition of Line

Each line shall be allocated a separate Line Number. The boundaries of a line shall be determined individually. Where a line is looped into a ground mounted substation this shall be treated as a boundary and separate Line Numbers allocated to each section of line. There is no necessity to change Line Number at a point of change of construction. Tee-offs from a line shall be allocated a separate Line Number. Tee-offs with several spurs may be given a single Line Number provided there is no interconnection with other parts of the HV System.

Lines may be divided into line parts. Where a line is not divided into parts there may be no more than 50 supports in the line.

3.2 Line Number

Lines will be numbered using a letter signifying the old Area set -up, followed by a Serial Number of no more than 4 digits.

The letters allocated to Areas are:-

Bradford	-	B
Huddersfield	-	H
Sheffield	-	S
Leeds	-	A
Wakefield	-	W
Hull	-	L
Grimsby	-	G

Allocation of line numbers must be agreed with the Control Centre.

3.3 Definition of Line Part

Lines may be divided into line parts. There shall be no more than 50 supports in any line part. Where a line embraces more than one type of construction or conductor size a change of line part must be made at the point of change. (This may be disregarded where a change involves no more than 2 spans).

3.4 Line Part Numbers

Line Part Numbers will be a two digit number (eg 01). Allocation of Part Numbers must be agreed with the Control Centre.

3.5 Definition of a Support

In the case of towers, H poles and the like, a support shall be taken to mean the complete structure.

3.6 Support Numbers

Each support within a line will be given an individual number. (For this purpose the Line Part will be ignored). This number will consist of no more than three numeric digits but may additionally include an alphabetic suffix.

3.7 Plated Number

The number shown on the number plate on an individual pole shall be known as the Plated Number and shall consist of Line Number plus Support Number. For HV lines the Line Part Number shall not form part of the Plated Number. The Plated Number will be in the form B163.

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The number plates to be used are shown on Drawing No Y003X3506.

3.8 Numbering Procedure for HV Lines

3.8.1 Construction of New Lines

The line will be given a new Line Number (and Part Numbers if appropriate), and the supports within that line will be consecutively numbered from one end, normally starting from No 1 in the new Line Number series as in Fig. 1.(see also sections 4.5 2 and 4.7)

3.8.2 Extension of Existing Lines

One of the following systems shall be adopted:-

- a) A new Line Number may be allocated and numbered as in 3.8.1 (see Fig 2.1).
- b) The new supports may be consecutively numbered in the old Line Number series, the serial numbers ascending from the pick-up point of the old line. (See Fig 2.2). However a new Line Part Number should be allocated.

3.8.3 Tee-offs from Existing Line

The tee will be allocated a new Line Number, and supports within that line will be consecutively numbered as in 3.8.1, commencing in the new Line Number series at the first support from the tee-off point (see Fig 3).

3.8.4 Interconnection of Existing Lines

3.8.4.1 Two Tees

If the interconnection is made by a tee at each end, the interconnector will be given a new Line Number and treated as in 3.8.3 (see Fig 4.1).

3.8.4.2 One Tee

If the interconnection is made by extending one line to tee into the other line, this will be treated as an extension of the former line as in 3.8.2 (see Fig 4.2).

3.8.4.3 Straight Interconnection

If the interconnection is made by extending both lines, this will be treated as an extension of one or other of the existing lines as in 3.8.2 (see Fig 4.3).

3.8.5 Looping-in of an Existing Line

Depending upon the length of the loop-in and its position within the existing line, one of the following methods will be adopted:-

3.8.5.1 Loop in Higher Numbered Section of Line

Where the loop-in is within the higher numbered half of the existing line, one leg of the loop-in will be treated as an extension of the lower numbered section of the existing line, and the other leg plus the higher numbered section of the existing line will be allocated a new Line Number.

However, a new support will not be allocated a Plated Number which was previously carried by an existing support. However, suffix A may be added to existing Support Number (see Fig 5.1).

3.8.5.2 Loop in Lower Numbered Section of Line

- a) Where the loop-in is within the lower numbered half of the existing line, and there are as many or more supports in the section of line between the lowest number and the loop-in point as there are in the loop-in, the lower numbered section of the existing line and its associated leg of the loop-in will be allocated a new Line Number. The other leg of the loop-in will be numbered in descending order from the point where it joins the existing line. A new support will not be allocated a Plated Number which was previously carried by an existing support. However suffix A may be added to existing Support Numbers (see Fig 5.2).
- b) If there are less supports between the lowest number in the existing line and the loop-in point than there are in the loop-in, the case will be treated as in 3.8.5.1 (see Fig 5.3).

In all cases where a ground mounted substation is looped into an overhead line it will be necessary to renumber one or both sections of the line.

3.9 Diversions**3.9.1 Equal Number of Supports**

Where the number of supports in the new position is equal to those in the existing position, the existing numbers will be transferred to the new supports, maintaining the correct sequence of numbers (see Fig 6.1).

3.9.2 Increased Number of Supports

Where the number of supports in the new position is more than those in the old position, the numbers will be transferred from the old to the new supports, and supplemented by numbers in the same line number series suffixed 'A' (see Fig 6.2).

3.9.3 Decreased Number of Supports

In the rare cases where the number of supports in the new position is less than those in the old position, the numbers will be transferred from the old to the new supports, a gap being left in the sequence (see Fig 6.3).

NOTE: In all the above three cases, it must be ensured that Pole History Sheets are amended.

3.10 Double Circuit Lines on Common Supports

Where two lines carried on separate supports converge to become a double circuit line on common supports, the common support must not bear two differing Support Numbers within the line series.

The common support must only bear one Support Number, although, in addition to the coloured symbols required to comply with Code of Practice OPS/102/028, each limb will carry different Line Numbers.

This can be achieved by either:-

- a) Ensuring that the serial numbers of supports in each line coincide at the common support.(see Fig 7.1(a) and 7.1(b)) or :
- b) Allocating separate Line Numbers to each section of the circuit concerned (see Fig 7.2.)

3.11 Additional Support erected in the run of an Existing Line

Where an additional support is erected in the run of an existing line it will be allocated the same number as the lower numbered adjacent pole, with the addition of a Suffix 'A' (see Fig 8).

3.12 Numbering System for LV Lines

3.12.1 Definition of Line

All LV lines supplied from a pole equipment or substation shall normally be treated as one line. However it is permissible to ignore normal open points and to treat a line interconnecting two substations as being all part of the same line. In this case both sections of the line will be numbered as if they were fed from one of the substations.

A line may be divided into line parts. Where a line is not divided into parts there will be no more than 50 supports in any line.

Note:

Historically in Hull Area all LV lines in a Parish have been allocated the same line number, however all new lines are being numbered to this Code of Practice.

3.12.2 Line Number

This shall consist of a prefix denoting the old Area plus a number of normally no more than 4 digits but exceptionally 5 digits. The local substation number will form the basis of this number.

The letters allocated to Area are:-

Bradford	-	C
Huddersfield District	-	J
Halifax District	-	K
Sheffield	-	T
Leeds	-	N
Wakefield	-	Y
Hull	-	M
Hull (Parish Numbers)	-	P
Grimsby	-	F

3.12.3 Definition of Line Part

Lines may be divided into line parts. There shall be no more than 50 supports in any line part.

3.12.4 Line Part Numbers

Line Part Numbers will be a two digit number (eg 02). Where a line is not divided into Parts the whole line shall be considered to be part 01.

3.12.5 Definition of Support

A support shall be taken to mean the complete structure. A support carrying both HV and LV conductors shall be numbered as an HV support only. Supports on service lines should be numbered. An underground cable termination onto surface wiring shall be classed as a support and allocated a number. A wall bracket will be allocated a support number only when it is directly connected to the LV mains (ie. wallbrackets only carrying surface wiring from building to building will not be given a support number).

3.12.6 Support Number

3.12.6.1 Individual Number

Each support within a Line Part will be given an individual number. This number will consist of no more than 2 numeric digits but may also include an alphabetic suffix.

3.12.6.2 Service Support

A service support may be given the same number as the line support to which it is connected followed by a suffix.

3.12.6.3 Wall Bracket or Wall Box

A wall bracket or wall box may be given the same number as the line support to which it is connected followed by a suffix. It is essential that wall brackets and wall boxes are given the appropriate suffix for identification purposes.

3.12.6.4 Support Number Suffixes

In Line Support	A B C
Service Support	D E F G H J K L
Wall Bracket	M N P R S T
Wall Box	V W X Y Z

3.13 Plated Number**3.13.1 Form of Number**

The number shown on the number plate on a individual pole shall be known as the Plated Number. The Plated Number shall include Line Number, Part Number and Support Number in the form:-

Line Number
Part Number - Support Number

A typical example being: C1234
02 - 21

3.13.2 LV Line Part Numbers

It should be noted that for LV lines the Line Part Number forms an integral part of the Plated Number.

3.13.3 Number Plates

The number plates to be used are shown on Drawing No Y003X3506 and shall be fitted to all wood, steel or concrete supports.

Although wall boxes and brackets must be numbered on mains records, numbering on site is optional. When it is desired to fit a number plate a smaller plate may be used.

3.14 Numbering Procedure for LV Lines**3.14.1 Consecutive Numbers**

As far as is practicable support numbers on any section of line shall be consecutive. If an extra support is added into an existing line a suffix "A" etc may be added to the support number.

3.14.2 Spurs

Each spur line shall normally have a separate series of numbers. If desired a spur line may be designated as a separate line part.

3.14.3 Examples

An example of a typical numbering scheme for LV lines is shown in fig 9 and for undereaves wiring connected to an underground system in fig 10.

3.14.4 Major Feeder Reorganisation

If a major reorganisation of feeding arrangements is made it may be necessary to re-number supports.

3.15 Mains Records

The Line Number, Part Number, if applicable, and Support Number will be recorded on Mains Records in accordance with Engineering Memorandum 176 - Underground and Overhead Mains Records. In addition the Field Operations Manager will notify the Consents Manager whenever support numbers on a line are changed to permit the amendment of Wayleave Payment Slips, and wayleave plans.

3.16 Control Diagrams

Line numbers and support numbers will be displayed on the Control DMS diagram at high voltage overhead line switchgear positions, and will also be shown on the detailed diagrams held in the Control Centre.

4. References

4.1 Internal

Code of Practice OPS/102/028

5. Definitions

None

EXPLANATORY NOTE (not forming part of this Code of Practice)

The changes from Operation Standard No.1 issued December 1978 are to the format of the document only

The changes introduced by the issue of December 1978 from the issue of August 1976 were as follows:

- 1) Requirement to change line number where construction changes have been removed.*
- 2) Line Part Numbers have been introduced.*
- 3) Definition of Plated Number introduced.*
- 4) System of renumbering supports for looping in new substation changed to facilitate computer re-referencing.*
- 5) Section on numbering of LV lines added.*
- 6) Requirement for separate line numbers on each side of ground mounted substation when hv line is looped in has been added.*

Appendices

Figure 1 New Construction

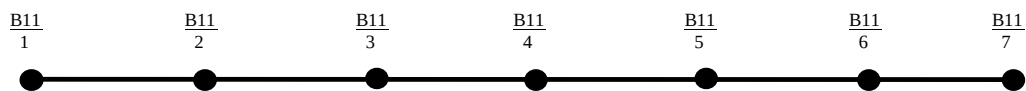


Figure 2.1 Extension of Existing Line

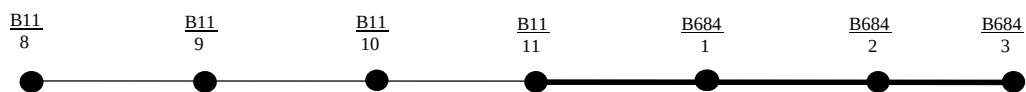


Figure 2.2 Extension of Existing Line

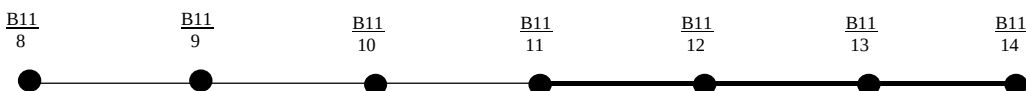


Figure 3 Tee-off Existing Line

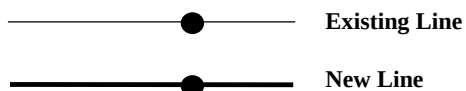
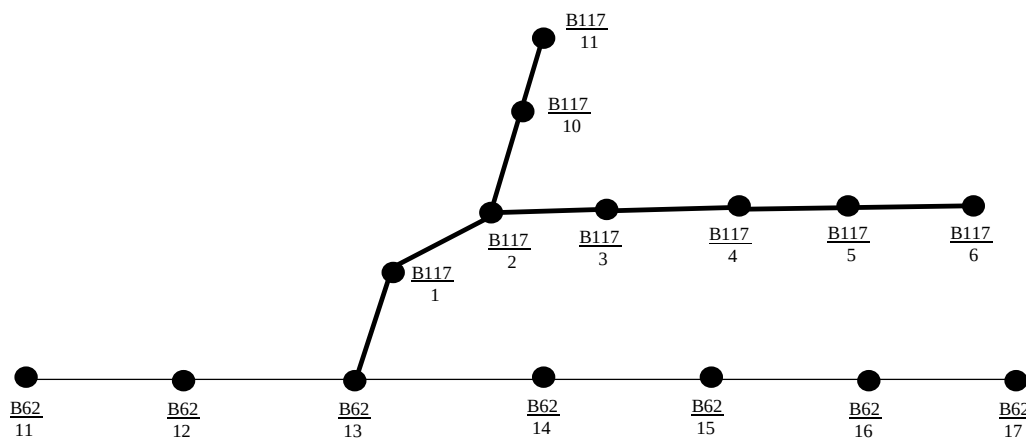


Figure 4 Interconnection of Existing Lines
Figure 4.1

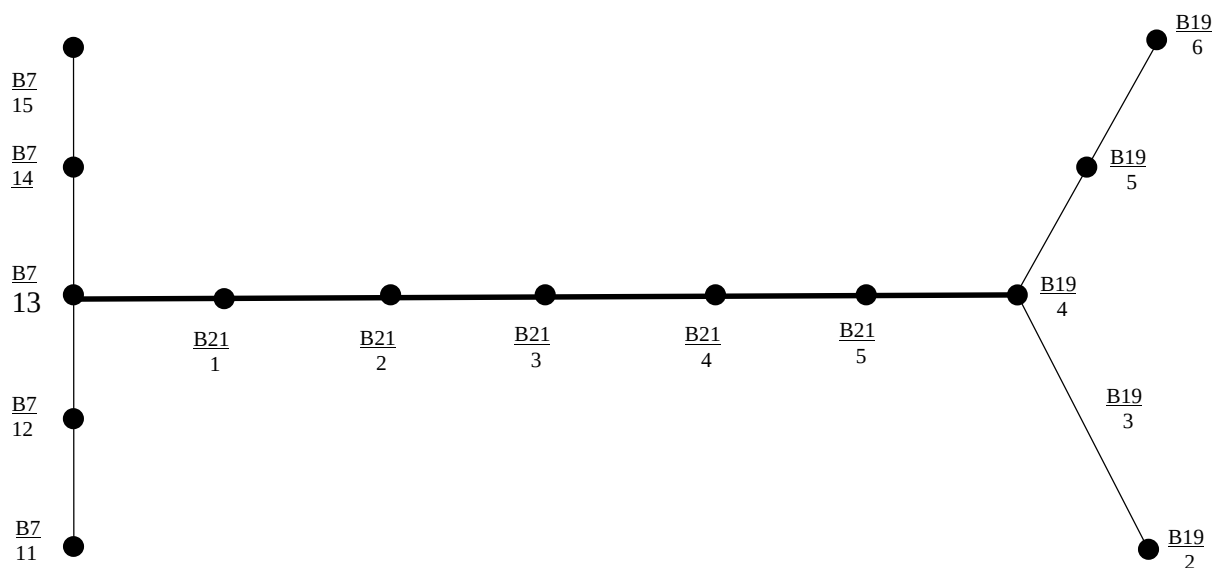


Figure 4.2

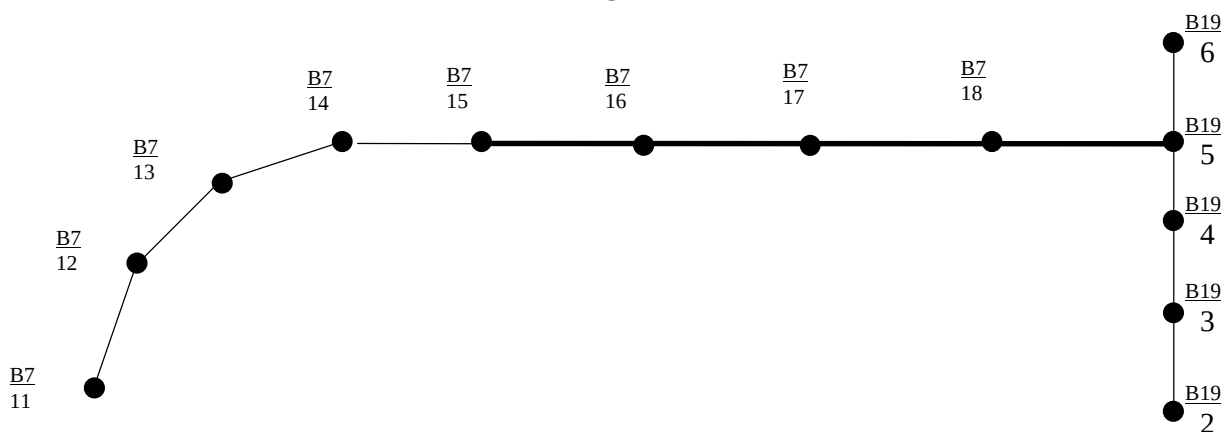
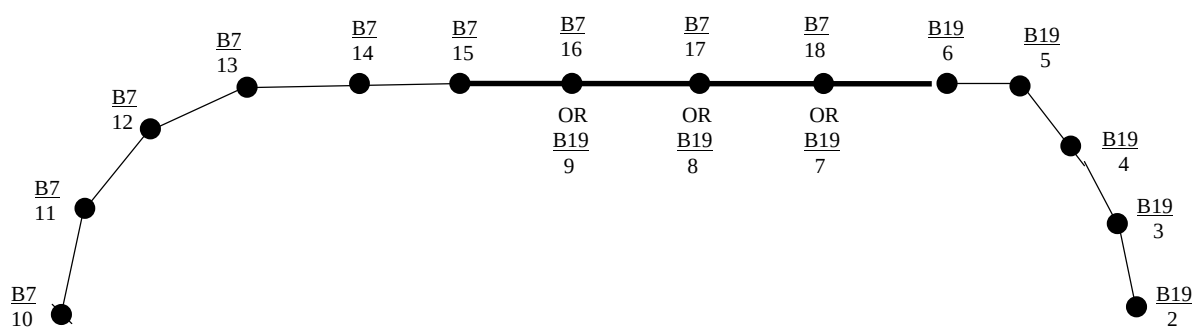


Figure 4.3



—●— Existing Line
—●— New Line

Figure 5 Looping into Existing Line

Figure 5.1

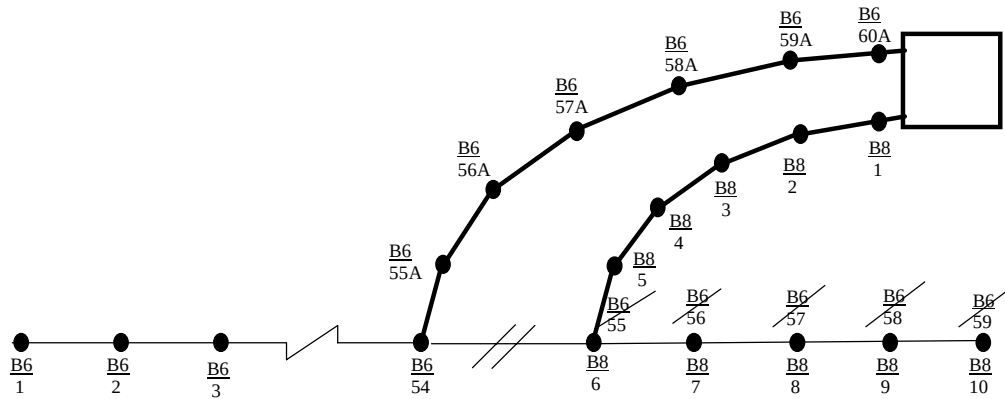


Figure 5.2

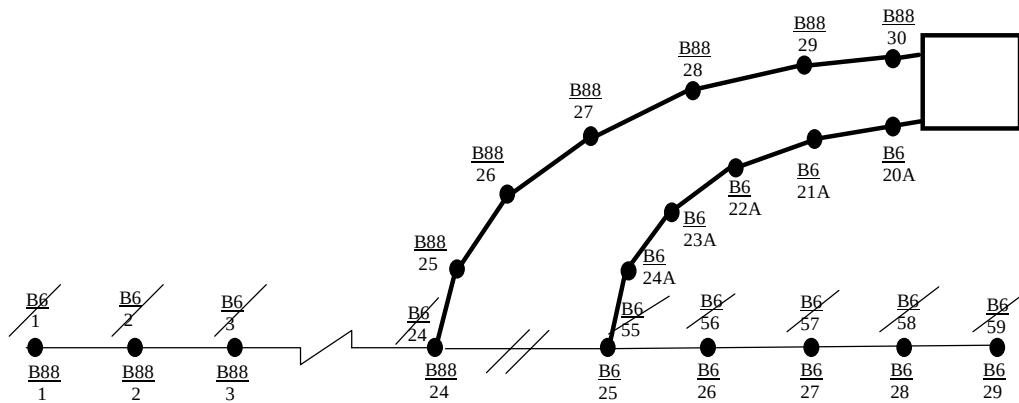
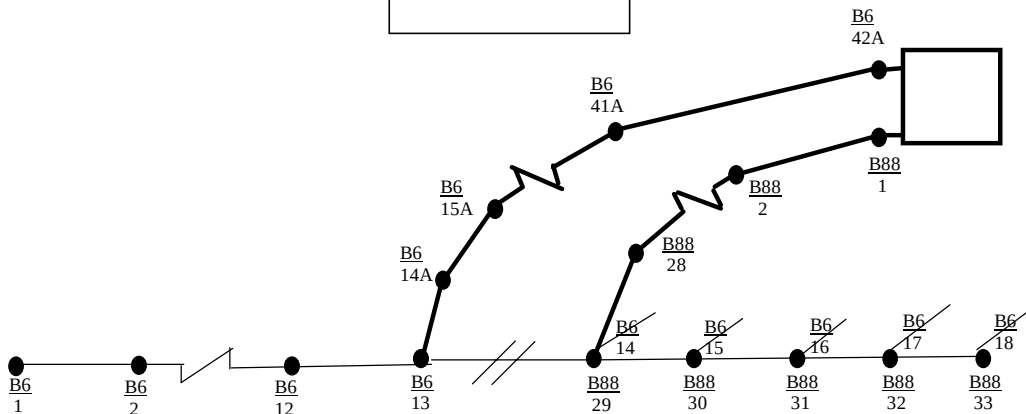


Figure 5.3



—●— Existing Line
 —●— New Line

Figure 6 Diversion of an Existing Line

Figure 6.1

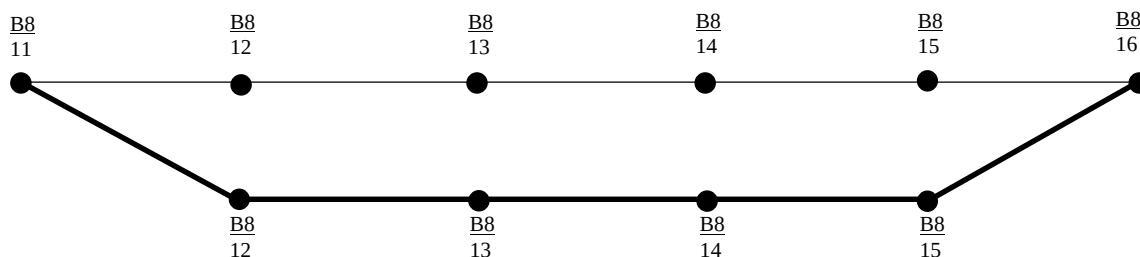


Figure 6.2

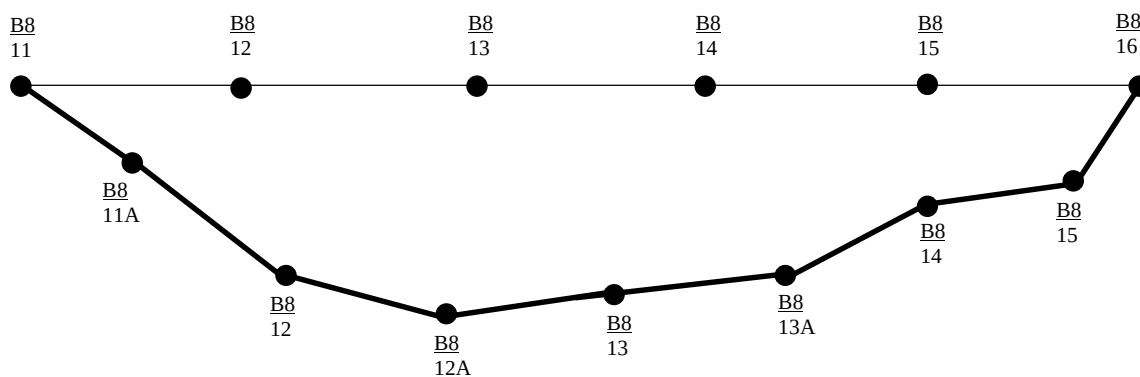


Figure 6.3

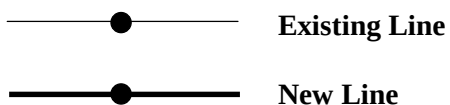
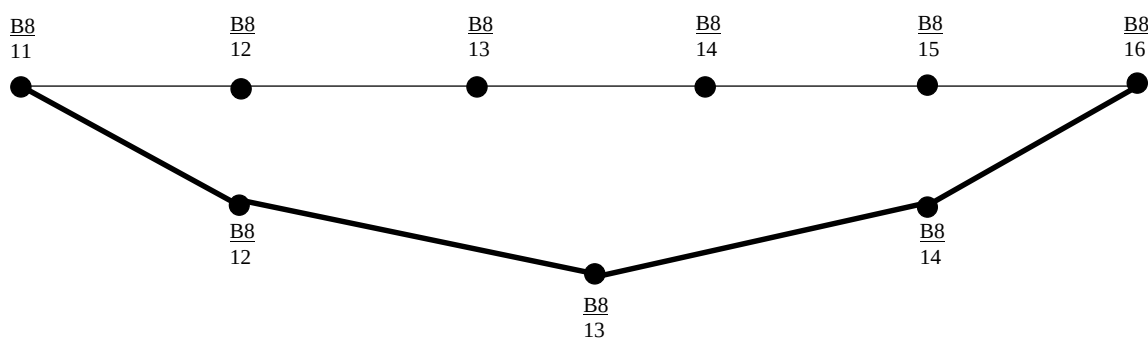


Figure 7 Double Circuit Lines on Common Supports

Figure 7.1A

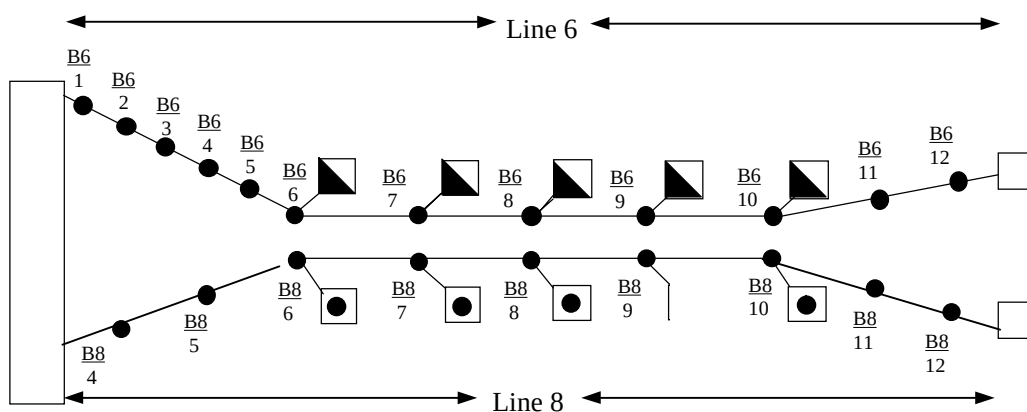


Figure 7.1B

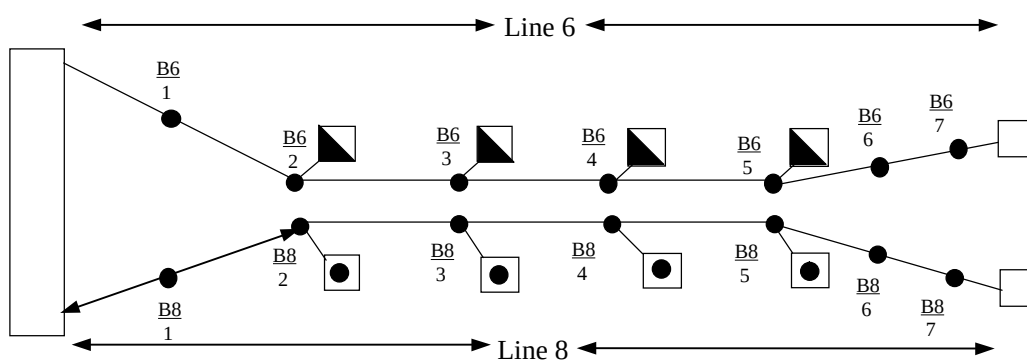
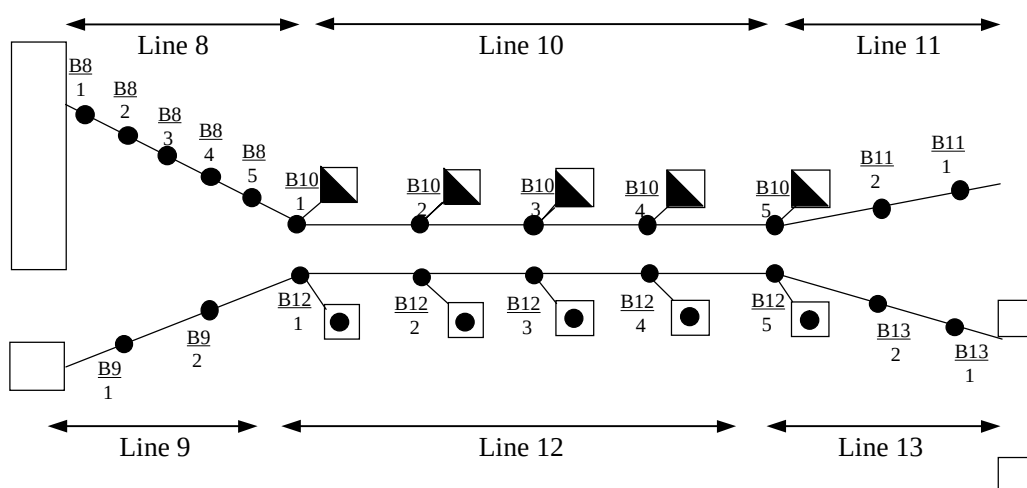


Figure 7.2



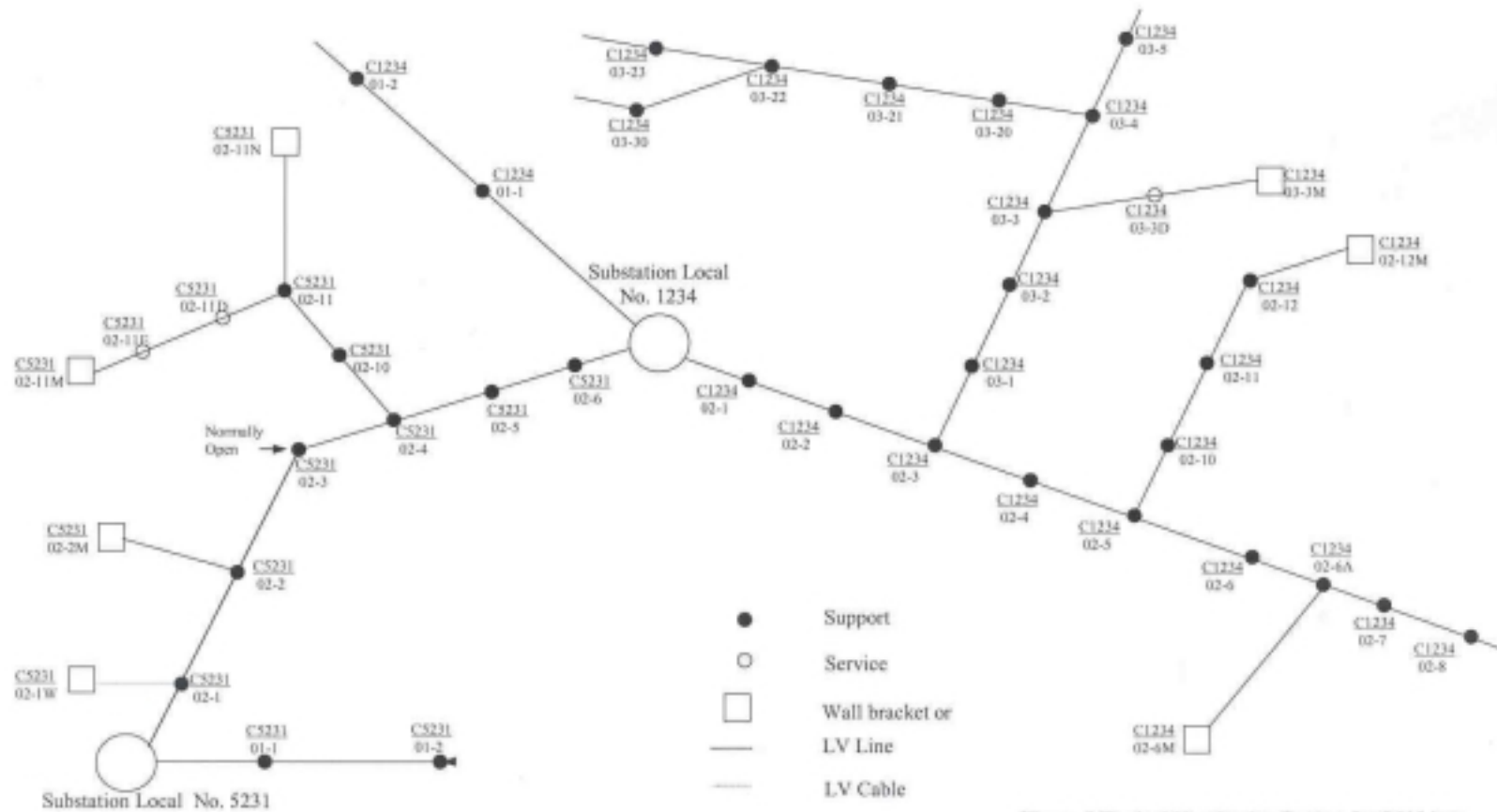


Figure 9 Typical Numbering System for LV Lines

Figure 10 Typical Numbering System for Wall Boxes Connected to Underground Cable.

